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SECTION:CSE-E

Importing the libraries nd showing the dataset:

```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler
data={
    'Student':['A', 'B', 'C', 'D','E','E' ],
    'Attendance':[85,90,np.nan,105,75,75],
    'Marks':[78,82,75,88,92,92]
}
df=pd.DataFrame(data)
print(df)
```

0.0s

```
  Student  Attendance  Marks
0       A        85.0     78
1       B        90.0     82
2       C         NaN     75
3       D       105.0     88
4       E        75.0     92
5       E        75.0     92
```

Identifying the invlid data and droping

```python
invalid=df[(df["Attendance"]<0) | (df['Attendance']>100)]
print(invalid)
df=df.drop(invalid.index)
```

0.0s Python

```
  Student  Attendance  Marks
3       D       105.0     88
```

Indentifying the duplicates and removing the duplictes value:

```
print("duplicated rows",df.duplicated().sum())
```

[48] 0.0s

```
duplicated rows 1
```

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```
df.drop_duplicates(inplace=True)
print(df)
```

[49] 0.0s

```
  Student  Attendance  Marks
0       A        85.0     78
1       B        90.0     82
2       C        85.0     75
3       D       105.0     88
4       E        75.0     92
```

Scaling the value through Minmax and Standardization:

```
scaler = MinMaxScaler()
df['marks_normalized']=scaler.fit_transform(df[['Marks']])
print("after min max",df['marks_normalized'])
```

0.1s Python

```
after min max 0    0.176471
1    0.411765
2    0.000000
4    1.000000
Name: marks_normalized, dtype: float64
```

```
sscaler=StandardScaler()
df['marks_Standardized']=sscaler.fit_transform(df[['Marks']])
print("after standardization",df['marks_Standardized'])
```

0.0s Python

```
after standardization 0   -0.584317
1    0.038954
2   -1.051771
4    1.597133
Name: marks_Standardized, dtype: float64
```

Indentifying the null vaues and filling:

```
print("null values",df.isnull().sum())
```

✓ 0.0s Python

```
null values Student       0
Attendance    1
Marks         0
dtype: int64
```

```
df['Attendance']=df['Attendance'].fillna(df['Attendance'].median())
print("after filling null values\n",df)
```

✓ 0.0s Python

```
after filling null values
   Student  Attendance  Marks  marks_normalized  marks_Standardized
0        A        85.0     78          0.176471           -0.798596
1        B        90.0     82          0.411765           -0.159719
2        C        86.0     75          0.000000           -1.277753
3        D       105.0     88          0.764706            0.798596
4        E        75.0     92          1.000000            1.437472
```

Final dataset:

```
print("cleaned datsaset\n",df
)
```

✓ 0.0s

```
cleaned datsaset
   Student  Attendance  Marks  marks_normalized  marks_Standardized
0        A        85.0     78          0.176471           -0.584317
1        B        90.0     82          0.411765            0.038954
2        C        85.0     75          0.000000           -1.051771
4        E        75.0     92          1.000000            1.597133
```

Data Cleaning techniques used:

Missing value:filling the missing value through median
Duplicates:droping the duplicates
Invalid:indentfying through **or** gate and droping

Normalization:scales data to fixed range,usually 0-1,all features have equal importance
Standardization:transform data to have the mean as zero,standard deviation to 1